

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2023 - 2024)

JEE(Advanced)**PRE-SCORE****29-11-2023****JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)**

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY**GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

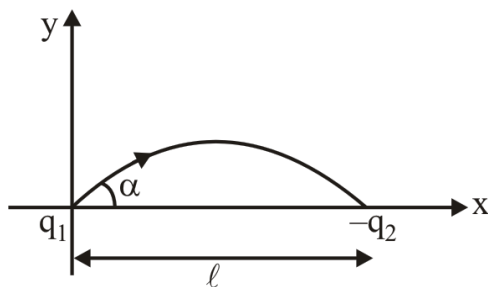
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. A leaky capacitor is one which does not have a perfectly insulating dielectric layer between the plates upon which the charge is stored. Such a capacitor contains a dielectric material filling the space between the plates, and with an effective resistivity of $\rho = \frac{5 \times 10^{12}}{8.85} \Omega\text{m}$. The area of the plates is $A = 1\text{m}^2$, and the dielectric film thickness $d = 8.85 \mu\text{m}$. The capacitor is charged from a 24 V supply connected in series with a $5 \text{ M}\Omega$ resistor. Dielectric constant of the material is 5.
- (A) The maximum charge on the capacitor is $60\mu\text{C}$.
- (B) The maximum charge on the capacitor is $120\mu\text{C}$.
- (C) If the capacitor is disconnected from the circuit, the time in which the charge on capacitor is halved is 25 sec.
- (D) If the capacitor is disconnected from the circuit, the time in which the charge on capacitor is halved is 50 sec.

2. An electric field line is generated at charge q_1 and terminates at q_2 as shown. The equation of the line can be

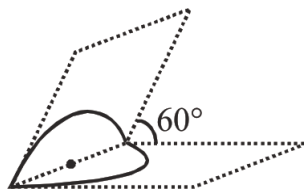


- (A) $q_1 \left(1 - \frac{x}{\sqrt{x^2 + y^2}} \right) + q_2 \left(1 - \frac{\ell - x}{\sqrt{(\ell - x)^2 + y^2}} \right) = q_1 (1 - \cos \alpha)$
- (B) $q_1 \left(1 - \frac{x}{\sqrt{x^2 + y^2}} \right) - q_2 \left(1 - \frac{\ell - x}{\sqrt{(\ell - x)^2 + y^2}} \right) = q_1 (1 - \cos \alpha)$
- (C) $q_1 \left(1 + \frac{x}{\sqrt{x^2 + y^2}} \right) + q_2 \left(1 - \frac{\ell - x}{\sqrt{(\ell - x)^2 + y^2}} \right) = q_1 (1 - \cos \alpha)$
- (D) $q_1 \left(1 + \frac{x}{\sqrt{x^2 + y^2}} \right) - q_2 \left(1 - \frac{\ell - x}{\sqrt{(\ell - x)^2 + y^2}} \right) = q_1 (1 - \cos \alpha)$

3. With what angular velocity the earth should spin in order that a body lying at 30° latitude may become weightless [R is radius of earth and g is acceleration due to gravity on the surface of earth]

- (A) $\sqrt{\frac{2g}{3R}}$ (B) $\sqrt{\frac{4g}{R}}$ (C) $\sqrt{\frac{g}{3R}}$ (D) $\sqrt{\frac{4g}{3R}}$

4. A uniformly charged ring of charge Q and radius R is folded from its diameter such that two halves make an angle 60° with each other then net electric field at centre of ring (O) is equal to



- (A) $\frac{\sqrt{3}kQ}{\pi R^2}$ (B) $\frac{2\sqrt{3}kQ}{\pi R^2}$
- (C) $\frac{2kQ}{\pi R^2}$ (D) $\frac{4kQ}{\pi R^2}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

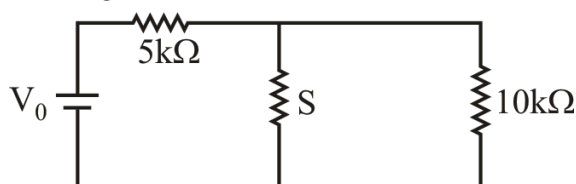
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

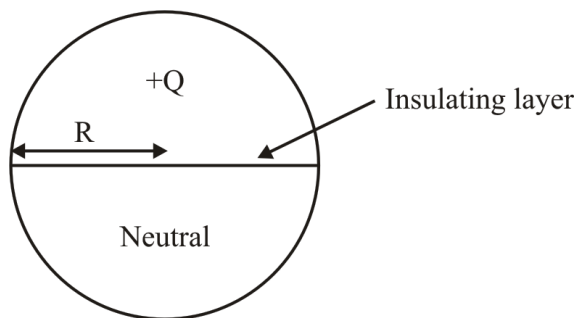
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Circuits with non-linear components are of great importance. The circuit is designed to prevent damage to the device represented by the $10\text{ k}\Omega$ load resistor when the voltage across the supply exceeds 2000 V . The component S has a V-I characteristic given by $I = kV^2$, where V is the voltage across S when current I flows through it. The constant k has a value $k = 1.0 \times 10^{-7}\text{ A V}^{-2}$.

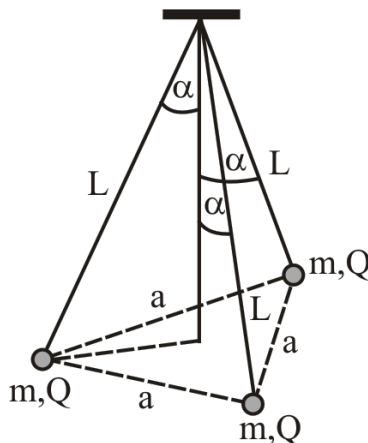


- (A) The current through the load resistor when operating normally under a supply of 2000 V is 0.1 A .
- (B) The current through the suppressor when operating normally under a supply of 2000 V is 0.2 A .
- (C) The supply voltage at which current through the load resistor doubles from its normal operating value is 3000 V .
- (D) The supply voltage at which current through the load resistor doubles from its normal operating value is 5000 V .

6. There is a solid metallic sphere of radius R , which is cut into two identical hemispheres. The cut surface is coated with a thin insulating layer of thickness d , and the two parts are put together so that the original shape of the sphere is restored. Initially, the sphere is electrically neutral. Then one of the hemispheres is given a positive charge $+Q$ while the other one remains neutral. You can assume $d \ll R$ in this problem.

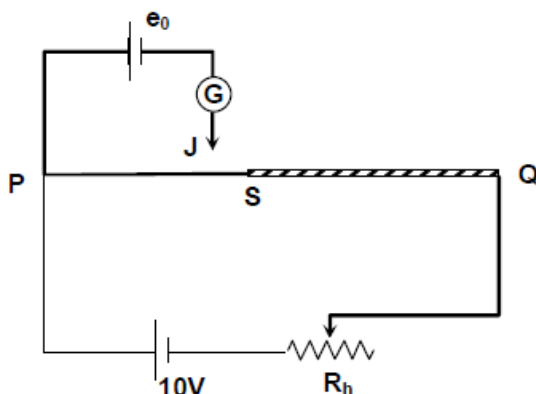


- (A) The surface charge density on the curved surface of the hemisphere is $\frac{Q}{4\pi R^2}$.
- (B) The electrostatic force on attraction between the two hemispheres is $\frac{3Q^2}{32\epsilon_0\pi R^2}$.
- (C) The electrostatic force of attraction between the two hemispheres is $\frac{3Q^2}{16\epsilon_0\pi R^2}$.
- (D) The electrostatic energy of the sphere is $\frac{Q^2}{2\pi\epsilon_0 R}$
7. Three point masses having charge $q = 3 \times 10^{-7}$ C and $m = 9 \times 10^{-4}$ kg are connected to three strings each of length 20 cm such that the other end of the strings is connected to a common point on the ceiling as shown in the diagram. Initially string make an angle $\alpha = 30^\circ$ with the vertical. System is released from rest.

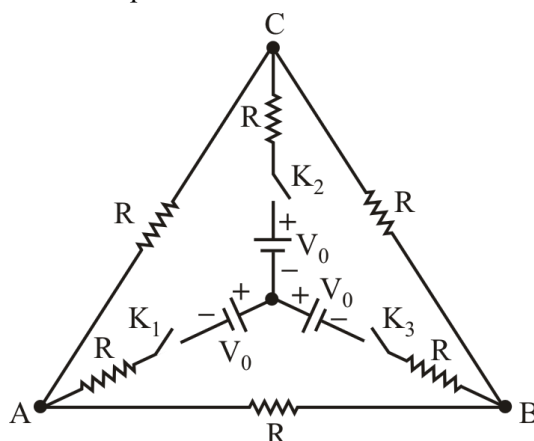


- (A) The angle made by the strings when the speed of point objects is maximum is 60° .
- (B) The maximum speed attained by the point objects is $\sqrt{2(\sqrt{3} + 1)} \text{ m/s}$.
- (C) The maximum angle made by the strings with the vertical is 90° .
- (D) The maximum tangential acceleration of point objects is 40 m/s^2 .

8. A potentiometer wire PQ is made by using two wires of different cross sectional area A and $2A$. The material of wire PS and SQ is same and their lengths are 5m each. A source battery of 10V and rheostat R_h are connected in the circuit as shown in the figure. Choose the correct option(s) regarding the balance point.

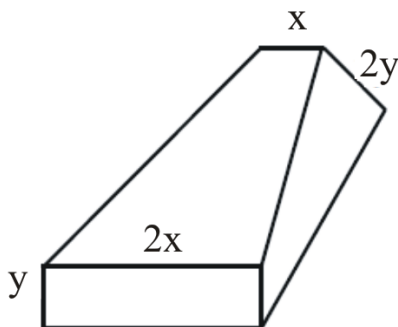


- (A) If a cell of emf $e_0 = 5\text{V}$ is used then balance point from the end P is at 7.5 m
 (B) If a cell of emf $e_0 = 5\text{V}$ is used then balance point from the end P is at 3.75 m
 (C) If a cell of emf $e_0 = 8\text{V}$ is used then balance point from the end P is at 7 m
 (D) If a cell of emf $e_0 = 8\text{V}$ is used then balance point from the end P is at 8.5 m
9. Initially all the switch in the circuit are open.



- (A) The current in AB after closing K_1 is $\frac{V_0}{2R}$.
 (B) The current in AC after closing K_1 and K_2 is $\frac{V_0}{2R}$.
 (C) The current in BC after closing all the switches is $\frac{V_0}{2R}$.
 (D) The current in AB after closing all the switches is $\frac{V_0}{R}$.

10. A conductor is made of an isotropic material (resistivity ρ) has rectangular cross-section. Horizontal dimension of the rectangle decreases linearly from $2x$ at one end to x at the other end and vertical dimension increases from y to $2y$ as shown in figure. Length of the conductor along the axis is equal to λ . A battery is connected across the ends of this conductor then



- (A) rate of generation of heat per unit length is maximum at the ends of conductor
- (B) rate of generation of heat per unit length is maximum at middle cross-section
- (C) drift velocity of conduction electrons is minimum at middle section
- (D) electric field intensity is maximum at ends of conductor.

Note : No questions have been asked in the section III of this paper. Please leave section III blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

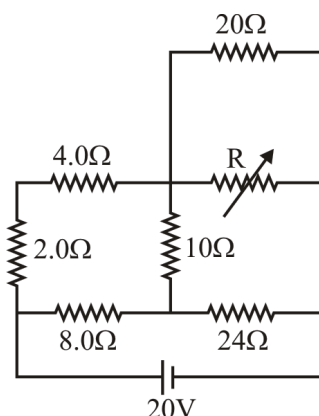
- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

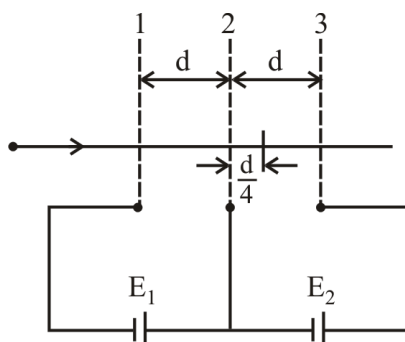
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. For the circuit shown in figure, what should be the value of the variable resistor R (in Ω) in order to minimise the power in the $10\ \Omega$ resistor. Fill the value of $\frac{R}{20}$.

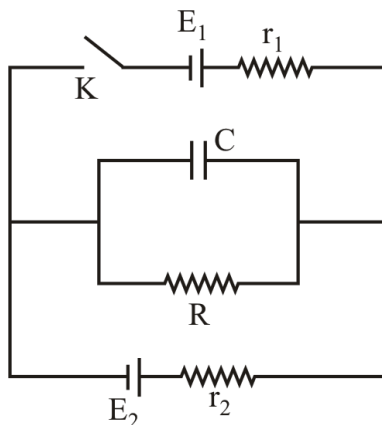


2. A charged particle having unity specific charge enters a system of three parallel meshes. The dimensions of meshes are very large as compared to the gap between the meshes. The meshes are connected to batteries of emf $E_1 = 15\text{ V}$ and $E_2 = 12\text{ V}$ as shown in the diagram.



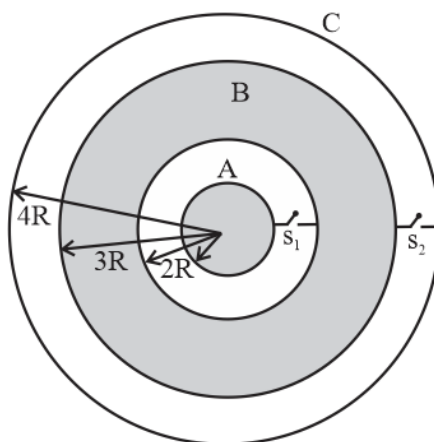
The charged particle comes to rest at a point $\frac{d}{4}$ behind the second mesh. The speed of particle at a large distance from the meshes is.

3. The key K is kept open for a very long time. The current through the capacitance C just after the key K is closed is $\frac{x}{10}$ A. The value of x is. Take : $E_1 = 40$ V, $r_1 = 200 \Omega$, $E_2 = 80$ V, $r_2 = 50 \Omega$ and $R = 150 \Omega$.



4. The time period of Jupiter is $(125)^{1/2}$ years. The distance of Jupiter from Sun in terms of astronomical unit is given by α AU. Find the value of α . (Astronomical unit is defined as the average distance of Earth from the Sun).

5.



Three concentric conducting spheres A (solid sphere of radius R), B (hollow sphere of inner radius $2R$ and outer radius $3R$) and C (hollow sphere of radius $4R$) are kept as shown. Sphere A is given charge Q . Capacity of the system is as follows:

C_1 : When both S_1 and S_2 are open

C_2 : When S_1 is closed and S_2 is open

C_3 : When S_1 is open and S_2 is closed

C_4 : When both S_1 and S_2 are close

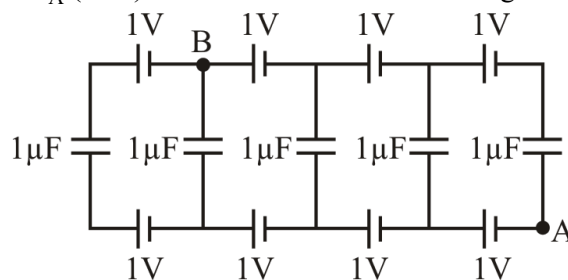
If $(C_3 + C_4) - (C_1 + C_2) = \frac{\alpha\beta\pi\epsilon_0 R}{15}$ (Where $\alpha\beta$ is a two digit number) then calculate $\beta - \alpha =$ _____.

6. A narrow tunnel is dug inside the earth along one of its diameters. Radius of earth is R and its mass is M .

A particle is projected inside the tunnel with velocity $\sqrt{\frac{2GM}{R}}$ from one of its ends then maximum velocity attained by the particle in the subsequent motion is $\sqrt{\frac{nGM}{R}}$ (assuming tunnel to be frictionless).

Find n .

7. The potential difference $V_B - V_A$ (in V) for the circuit shown in the figure is



8. Two conducting spheres of radii r and $2r$ are placed at very large separation. Each sphere possesses charge Q . These spheres are connected with a conducting wire of resistance R then current reduces to half the initial current after time $\frac{8\pi\epsilon_0 r R \ln 2}{n}$. Find n .

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

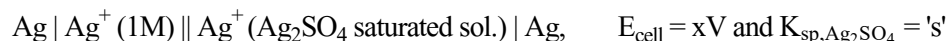
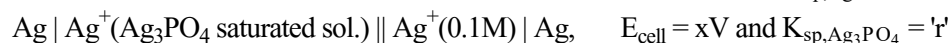
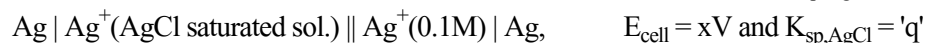
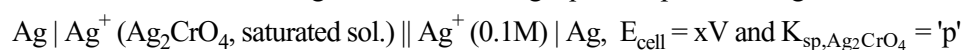
- This section contains **FOUR (04)** questions.
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

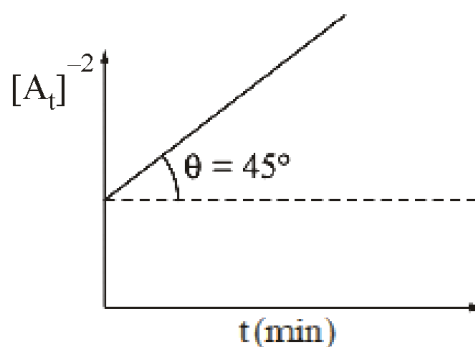
Negative Marks : -1 In all other cases

1. Consider the four different galvanic cells having equal cell potential in given conditions :



The correct option for solubility product for all the four cases is (by considering all cells are in equilibrium stage)

- (A) $s > q > p > r$ (B) $s < p < q < r$ (C) $r > s > q > p$ (D) $r > q > p > s$
2. For an electrolytic cell undergoing following balanced reaction :
- $$\text{NO}_3^- + 3\text{H}_3\text{O}^+ + \text{Hg}_2^{2+} \rightarrow 2\text{Hg}^{2+} + \text{HNO}_2 + 4\text{H}_2\text{O}$$
- If an average current of 10 amp. passes through the cell, how long will it take to produce 0.3 moles of HNO_2 by this reaction?
- (A) 1.6 hours (B) 16 hours (C) 3.2 hours (D) 32 hours
3. For the reaction $3\text{A} \rightarrow 2\text{B}$, the following graph is obtained experimentally. The rate of reaction when $[\text{A}] = 0.2 \text{ M}$, is



- (A) $8 \times 10^{-4} \text{ M min}^{-1}$ (B) $\frac{16}{6} \times 10^{-4} \text{ M min}^{-1}$
- (C) $\frac{4}{3} \times 10^{-3} \text{ M min}^{-1}$ (D) $\frac{16}{3} \times 10^{-3} \text{ M min}^{-1}$
4. Which of the following is colloid in which dispersed phase is gas and dispersion medium is liquid ?
- (A) Mist (B) Ruby Glass
- (C) Whipped Cream (D) Lava

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. NH_3 concentration in a saturated solution of AgBr(s) is maintained at 0.2M. The correct information(s) related with the solution is/are :

Consider the data given below

$$K_{\text{sp}} \text{ of Ag Br} = 2 \times 10^{-10}$$

$$K_{\text{f}} \text{ of Ag(NH}_3)_2^+ = 8 \times 10^6$$

$$\lambda_{(\text{Ag}^+)}^\circ = 6 \times 10^{-3} \text{ Sm}^2 \text{ mol}^{-1}$$

$$\lambda_{(\text{Br}^-)}^\circ = 8 \times 10^{-3} \text{ Sm}^2 \text{ mol}^{-1}$$

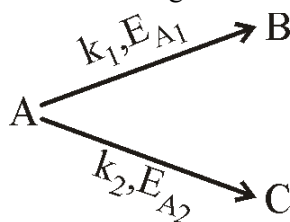
(A) Conductivity of $\text{Br}^- = 6.4 \times 10^{-2} \text{ S m}^{-1}$

(B) Conductivity of $\text{Ag}^+ = 1.5 \times 10^{-7} \text{ S m}^{-1}$

(C) $[\text{Ag}^+] = 8 \times 10^{-3} \text{ M}$

(D) $[\text{Br}^-] = 2 \times 10^{-3} \text{ M}$

6. EMF of the cell: $\text{Cd(s)} | \text{CdCl}_2 \cdot 5\text{H}_2\text{O (saturated solution)} | \text{AgCl(s)} | \text{Ag(s)}$ is +0.70 V at 0°C and +0.60 V at 50°C . If ΔH° and ΔS° are temperature independent, then the correct information(s) regarding the cell reaction is/are
- (A) $\Delta G^\circ = -115.8 \text{ kJ at } 50^\circ\text{C}$
- (B) $\Delta G^\circ = 135.1 \text{ kJ at } 0^\circ\text{C}$
- (C) $\Delta S^\circ = -386 \text{ J/K}$
- (D) $\Delta H^\circ = 221.18 \text{ kJ}$
7. A substance A reacts at 500 K by parallel reactions to give B and C as shown



Reaction -1 : $k_1 = 10^{-3} \text{ sec}^{-1}$, $E_{A1} = 80 \text{ kcal/mole}$

Reaction -2 : $k_2 = 10^{-2} \text{ sec}^{-1}$, $E_{A2} = 100 \text{ kcal/mole}$

Identify the **INCORRECT** statement(s).

- (A) The net activation energy for conversion of **A** is 90 Kcal/mole.
- (B) The pre-exponential factor for 1st reaction is smaller as compared to 2nd reaction.
- (C) $|\Delta H|$ for **reaction-1** must be greater than **reaction-2**.
- (D) Fraction of **A** reacting by either of the reaction will be same at 500 K.
8. For the reaction $\text{CHCl}_3 + \text{Cl}_2 \xrightarrow{h\nu} \text{CCl}_4 + \text{HCl}$, mechanism is given below :
- $$\text{Cl}_{2(g)} \xrightleftharpoons[k_{-1}]{k_1} 2\text{Cl}_{(g)}$$
- $$\text{Cl}_{(g)} + \text{CHCl}_{3(g)} \xrightarrow{k_2} \text{HCl}_{(g)} + \text{CCl}_{3(g)}$$
- $$\text{CCl}_{3(g)} + \text{Cl}_{(g)} \xrightarrow{k_3} \text{CCl}_{4(g)}$$
- $k_1 = 4.8 \times 10^3 \text{ sec}^{-1}$, $K_{-1} = 12 \times 10^3 \text{ M}^{-1} \text{ sec}^{-1}$, $k_2 = 1.3 \times 10^{-2} \text{ M}^{-1} \text{ sec}^{-1}$, $k_3 = 2.1 \times 10^2 \text{ M}^{-1} \text{ sec}^{-1}$
- Given that second step of the mechanism is slowest step.
- Correct statement is/are :
- (A) Order of reaction is 3/2
- (B) Magnitude of overall rate constant (conc. in M, time in sec) is 2.6×10^{-2}
- (C) If concentration of CHCl_3 is increased to four times rate of reaction increases by a factor of two
- (D) If concentration of Cl_2 is increased to four times rate of reaction increases by a factor of two

9. Which of the following statement(s) is/are incorrect with respect to enzyme catalysed reactions?
- (A) The cavities on enzyme “Maltase” & structure of molecules of Maltose have complimentary shapes for conversion of Maltose to glucose.
- (B) Just like other catalysts, catalytic action of enzyme is more at a lower temperature.
- (C) In presence of vitamins & some metal ions; enzyme activity generally increases.
- (D) Most of the enzyme catalysed reactions are elementary.
10. For the coagulation of a particular colloidal solution the order of coagulation value of electrolytes is as follows.



- (A) In electroosmosis of this colloidal solution dispersion medium move towards cathode.
- (B) Diffused layer of the colloidal sol particles contains positively charged ions.
- (C) It may be a metal sol such as Pt, Au or Hg sol
- (D) It may be basic dye sol such as methylene blue.

Note : No questions have been asked in the section III of this paper. Please leave section III blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

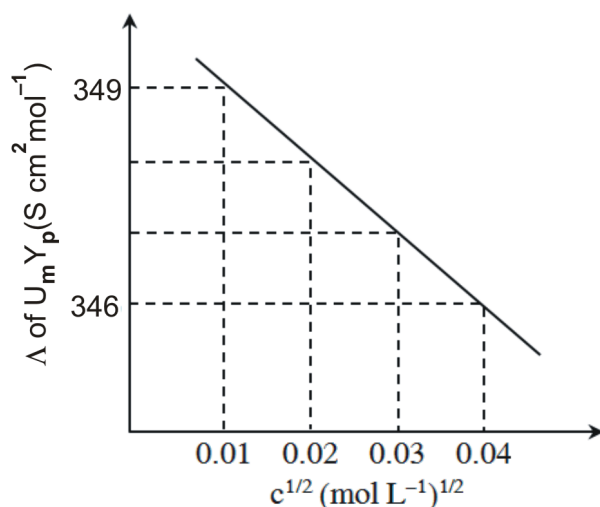
Negative Marks : -1 In all other cases

1. Consider the strong electrolytes Z_mX_n , U_mY_p and V_mX_n . Limiting molar conductivity (λ^0) of Z_mX_n and V_mX_n are 250 and 440 $S\text{ cm}^2\text{ mol}^{-1}$, respectively. The nearest integer value of $(m + n + p)$ is _____.
Given :

Ion	Z^{n+}	U^{p+}	V^{n+}	X^{m-}	Y^{m-}
$\lambda^0(S\text{ cm}^2\text{ mol}^{-1})$	50.0	25.0	100.0	80.0	100.0

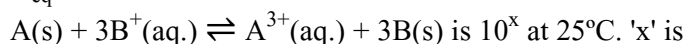
λ^0 is the limiting molar conductivity of ions

The plot of molar conductivity (Λ) of U_mY_p vs $c^{1/2}$ is given below.



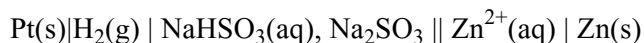
2. For the reaction : $2\text{NO}(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{N}_2\text{O}(\text{g}) + \text{H}_2\text{O}(\text{g})$, the value of $\frac{-dp}{dt}$ was found to be 1.5PaS^{-1} for a pressure of 372 Pa for NO and 0.25PaS^{-1} for a pressure of 152 Pa for NO, the pressure of H_2 being constant. If pressure of NO was kept constant, the value of $\frac{-dp}{dt}$ was found 1.60PaS^{-1} for a pressure of 289 Pa for H_2 and 0.79PaS^{-1} for a pressure of 144 Pa for H_2 . If order of reaction with respect to NO and H_2 are a and b, respectively, then find nearest integer value of $(a + b)$?

3. K_{eq} for the reaction :

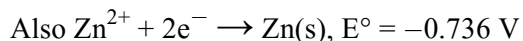


Given : $\frac{2.303RT}{F} = 0.06V$, $E_{A/A^{2+}}^0 = 0.25V$, $E_{A^{2+}/A^{3+}}^0 = 0.1V$, $E_{B/B^+}^0 = 0.1V$

4. The voltage of the cell is $-0.6769V$ at 25°C temperature.



$$1 \text{ atm} \quad 0.4M \quad 6.44 \times 10^{-3}M \quad 0.04M$$



If $K = \frac{[H^+][SO_3^{2-}]}{[HSO_3^-]} = 3.22 \times 10^{-x}$. Then find value of 'x' ?

Given : $\frac{2.303RT}{F} = 0.0591V$

5. $CH_3COOC_2H_5(aq) + H_2O(l) \xrightarrow{H^+} CH_3COOH(aq) + C_2H_5OH(aq)$

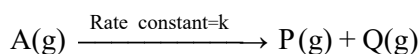
Rate = $K[H^+]^1[H_2O]^1[CH_3COOC_2H_5]^1$, $K = 1.8 \times 10^{-3} M^{-2} \text{sec}^{-1}$.

In an experiment 1 L of 1 M $CH_3COOC_2H_5$ is hydrolysed by mixing 1 L 0.2 M HCl. Immediately after starting the reaction, V mL of reaction mixture requires 10 mL 0.1 M NaOH, after long time of starting the reaction V mL of reaction mixture requires 100 ml of 0.1 M NaOH for complete neutralisation. If volume of 0.1 M NaOH required for complete neutralisation of V mL of reaction mixture after 230.3 sec is 'x' mL find nearest integer value of $\left(\frac{x}{10}\right)$?

6. Total number of sols among below given options which can be coagulated by Al^{+3} ion are?

Eosin, methylene blue, Congo red, TiO_2 , Ag Sol, Sb_2S_3 Sol, Al_2O_3 sol. Haemoglobin sol, Clay sol., Charcoal sol.

7. For a reaction



carried in closed and rigid container at constant temperature T, following informations are provided.

	Initial Concentration	$T_{1/2}$	$T_{3/4}$
A(g)	$1 \times 10^{-2}M$	10 Min	30 Min

If initial rate of reaction in mole/lit/min is represented in scientific notation as $x \times 10^{-y}$ then find the value of y.

8. 1 gm of charcoal having surface area $3.01 \times 10^2 m^2$ is mixed in 100 ml of 0.5 M acetic acid solution; after sufficient time concentration of solution becomes 0.4 M. The surface area occupied by one molecule of acetic acid is $y \times 10^{-20} m^2$. Then y is. [Given : $N_A = 6.02 \times 10^{23}$]

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Let A_1, A_2 and A_3 be the regions on $\mathbb{R}^2$ defined by
 $A_1 = \{(x, y) : x \geq 0, y \geq 0, 2x + 2y - x^2 - y^2 > 1 > x + y\}$,
 $A_2 = \{(x, y) : x \geq 0, y \geq 0, x + y > 1 > x^2 + y^2\}$,
 $A_3 = \{(x, y) : x \geq 0, y \geq 0, x + y > 1 > x^3 + y^3\}$
 Denote by $|A_1|, |A_2|$ and $|A_3|$ the areas of the regions A_1, A_2 and A_3 respectively. Then
- (A) $|A_1| > |A_2| > |A_3|$
 (B) $|A_1| > |A_3| > |A_2|$
 (C) $|A_1| = |A_2| < |A_3|$
 (D) $|A_1| = |A_3| > |A_2|$
2. Solution of the differential equation $\frac{\sqrt{x} dx + \sqrt{y} dy}{\sqrt{x} dx - \sqrt{y} dy} = \sqrt{\frac{y^3}{x^3}}$ is given by (Base of all log = e)
- (A) $\frac{3}{2} \log\left(\frac{y}{x}\right) + \log\left|\frac{x^{\frac{3}{2}} + y^{\frac{3}{2}}}{x^{\frac{3}{2}}}\right| + \tan^{-1}\left(\frac{y}{x}\right)^{\frac{3}{2}} + c = 0$
 (B) $\frac{2}{3} \log\left(\frac{y}{x}\right) + \log\left|\frac{x^{\frac{3}{2}} + y^{\frac{3}{2}}}{x^{\frac{3}{2}}}\right| + \tan^{-1}\left(\frac{y}{x}\right) + c = 0$
 (C) $\frac{2}{3} \log\left(\frac{y}{x}\right) + \log\left(\frac{x+y}{x}\right) + \tan^{-1}\left(\frac{y^{\frac{3}{2}}}{x^{\frac{3}{2}}}\right) + c = 0$
 (D) $\frac{1}{2} \log(x^3 + y^3) + \tan^{-1}\left(\frac{y}{x}\right)^{\frac{3}{2}} + c = 0$

3. Let $I_1 = \int_{-2}^2 \frac{x^6 + 3x^5 + 7x^4}{x^4 + 2} dx$ and $I_2 = \int_{-3}^1 \frac{2(x+1)^2 + 11(x+1) + 14}{(x+1)^4 + 2} dx$, then the value of $I_1 + I_2$ is

- (A) 8
- (B) 200/3
- (C) 100/3
- (D) None of these

4. $\int \frac{x (x \tan^{-1} x + (\ln x)(\ln(\ln x))) + \tan^{-1} x}{(x^3 + x) \ln x} dx = \phi(x)$ where $\phi(e) = 0$ then the value of $\lim_{x \rightarrow 1} \frac{\phi(x)}{\tan\left(\frac{\pi x}{2}\right)}$

is equal to

- (A) 0
- (B) $\frac{\pi}{4}$
- (C) $-\frac{\pi}{4}$
- (D) $-\frac{1}{2}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. For $0 < a < 1$, $-1 < x < 1$. Let $f(x) = \frac{a-x}{1-ax}$ and if $\int_0^a \frac{1-(f(x))^6}{1-x^2} dx = F(a)$ then

(A) $F(a)$ is increasing function

(B) $F(a)$ is an odd function

(C) $F(1) = \frac{23}{15}$

(D) $F(x) = 0$ has exactly one real root

6. Let $y = \sin x$ and $I = \int (x-y)\tan^2\left(\frac{x}{2}\right).d(x+y)$, then

(A) $d(y+x) = \cot^2\left(\frac{x}{2}\right).d(x-y)$

(B) $\frac{d^2x}{dy^2} = \frac{\sin x}{\cos^3 x}$

(C) $I = (x-y)^2 + C$

(D) $I = \frac{(x-y)^2}{2} + C$

7. The area bounded by the function $f(x) = |\cos x|$ and $g(x) = \sin x$ between the ordinates $x = 0$ and $x = n\pi$, $n \in \mathbb{N}$, is equal to
- (A) $4n\sqrt{2}$ if n is odd
 (B) $2n\sqrt{2}$ if n is even
 (C) $2\sqrt{2}(n+1)$ if n is odd
 (D) $2\sqrt{2}(n+1) - 4$ if n is odd
8. If the Area bounded by $y = (\sin x)^{\operatorname{cosec} x}$, $y = (\operatorname{cosec} x)^{\sin x}$, $y = (\sin x)^{\sin x}$ and $y = (\operatorname{cosec} x)^{\operatorname{cosec} x}$ between the ordinates $x = 0$, $x = \frac{\pi}{2}$ denoted by A_1, A_2, A_3 and A_4 then
- (A) A_4 is the greatest
 (B) A_1 is the least
 (C) $A_2 > A_3$
 (D) $A_2 < A_3$
9. Let $I_n = \int \frac{\cos^{2n} x}{\sin x} dx$ satisfies the recursive relation $(2n+1)I_{n+1} = (2n+1)I_n + \phi(x)$, if $\phi'(x) = f(n, x)$, then which is/are correct
- (A) $f\left(5, \frac{\pi}{2}\right) = 0$
 (B) $f\left(5, \frac{\pi}{4}\right) = \frac{-11}{32\sqrt{2}}$
 (C) $f(6, \pi)$ is even number
 (D) $f\left(6, \frac{\pi}{2}\right)$ is prime number
10. Which is/are correct ?
- (A) $\int_0^2 \frac{1}{2 + \arctan x} dx \geq \ln 2$
 (B) $\int_0^2 \frac{1}{2 + \arctan x} dx < \ln 2$
- (C) If $f(x)$ is such that $|f(u) - f(v)| \leq |u - v|$ then $\left| \int_a^b f(x) dx - (b-a) \cdot f(a) \right| \leq \frac{(b-a)^2}{2}, (a < b)$
 (D) If $f(x)$ is such that $|f(u) - f(v)| \leq |u - v|$ then $\left| \int_a^b f(x) dx - (b-a)f(a) \right| \geq \frac{(b-a)^2}{2}, (a < b)$

Note : No questions have been asked in the section III of this paper. Please leave section III blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. The value of $\int_{\frac{\pi}{4}}^{\frac{65\pi}{4}} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})}$ is $k\pi$, then the value of k is equal to
2. Let for $x \in \mathbb{R}$, $S_0(x) = x$, $S_k(x) = C_k x + k \int_0^x S_{k-1}(t) dt$, where $C_0 = 1$, $C_k = 1 - \int_0^1 S_{k-1}(x) dx$, $k = 1, 2, 3, \dots$
Then $\left[\frac{S_2(3) + 6C_3}{3} \right]$ is equal to (Where $[]$ denotes Greatest Integer Function)
3. If $\int \frac{3x^2 + 2x}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5} dx = F(x)$, then find the value of $[F(1) - F(0)]$, where $[.]$ represents greatest integer function
4. Let the functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$.
Then the area of the region bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 1$ is $2 - \sqrt{\lambda}$, then the value of λ is
5. Let a , b and λ be positive real numbers. Suppose P is an end point of the latus rectum of the parabola $y^2 = 4\lambda x$, and suppose the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the point P . If the tangents to the parabola and the ellipse at the point P are perpendicular to each other, then the eccentricity of the ellipse is e . If $\left(e, \frac{a}{b}\right)$ is a point on the ellipse $15x^2 + 3y^2 = K$, then K is equal to _____
6. Let $\frac{dy}{dx} = \frac{ax - by + a}{bx + cy + a}$, $a, b, c \in \mathbb{R}$ represents a circle with centre (α, β) then $\alpha + 2\beta =$
7. The area bounded by the curve $y = f(x) = \int_1^{1/x^3} \left(\frac{1}{1+t^4} + t^{-4/3} \right) dt - 3 \int_x^1 \frac{t^8}{1+t^{12}} dt$ and the co-ordinate axes is equal to $\frac{p}{q}$ (where $p, q \in \mathbb{N}$, G.C.D = 1) then the value of $p + q$ is
8. Let $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x + \cos x} \right)^2 dx$ and $v = \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{\cos x} \right)^2 dx$, then the value of $\frac{v}{u}$ equals

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
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E-24/24

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